



Lecture 14: Synthesis Solar System in Context, Astrobiology

Planetary Systems

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Learning Objectives

By the end of this lecture, you will be able to:

- Place our solar system within the observed exoplanet population
- Evaluate habitability as a coupled *systems* property, not a single condition
- Derive classical habitable-zone boundaries from stellar luminosity and liquid water
- Identify the most promising life-detection targets for the next decade

Habitability is a trajectory, not a snapshot.

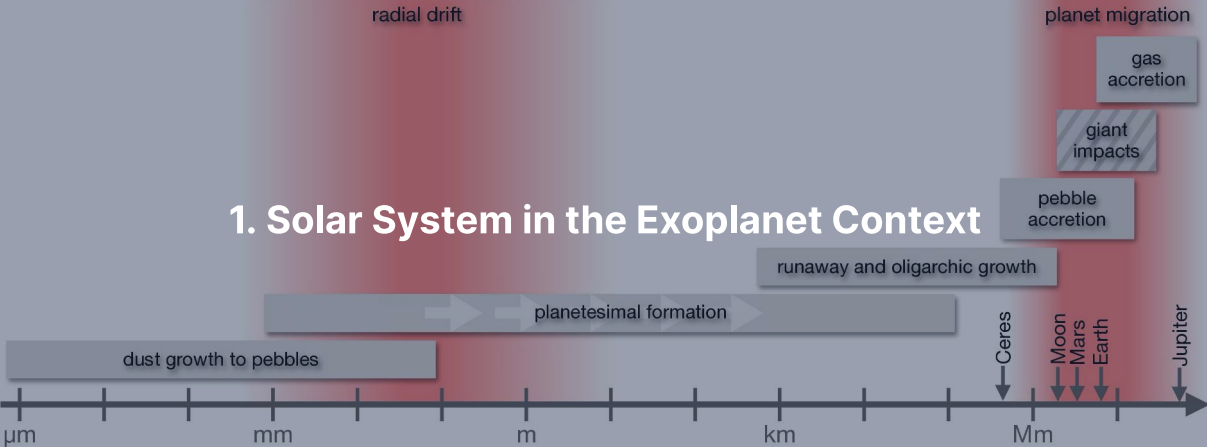
The Thread of the Course

The same physical laws govern planetary formation, evolution, and climate throughout the galaxy.

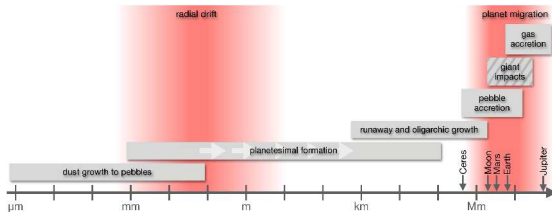
- L1–L8: accretion, differentiation, internal heat engines, and atmosphere-surface coupling
- L9–L12: terrestrial and giant planet evolution traced body by body
- L13: over 6000 exoplanets revealing architectural diversity

Today: Step back to evaluate the solar system in context and test astrobiological habitability.

1. Solar System in the Exoplanet Context

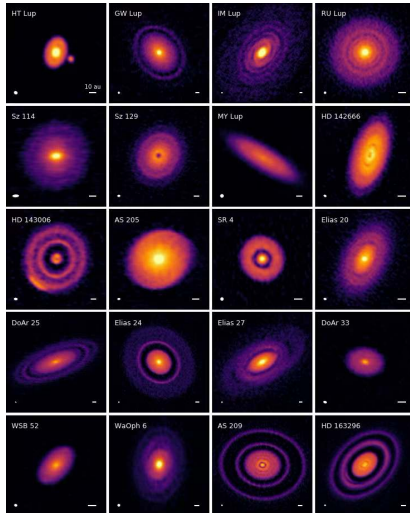


Formation Theory Meets Observation

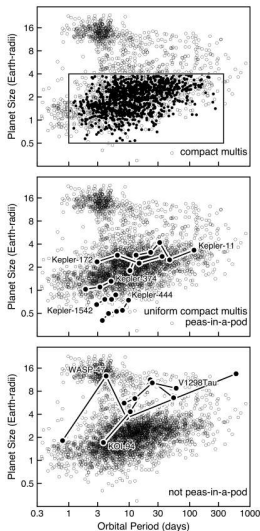


- Dust $\rightarrow$ pebbles $\rightarrow$ planetesimals (km-scale bodies) $\rightarrow$ embryos.
- Streaming instability (pebble clumping) + pebble accretion + runaway envelope accretion.
- Disk lifetime 3–5 Myr is the deadline.
- ALMA disk substructure is direct observational input.

ALMA's DSHARP Reveals Forming-Planet Signatures



- 240 GHz continuum, 20 nearby disks.
- Concentric rings, gaps, asymmetries: nearly ubiquitous.
- Most are interpreted as signatures of in-progress planet growth.
- Andrews+ 2018.



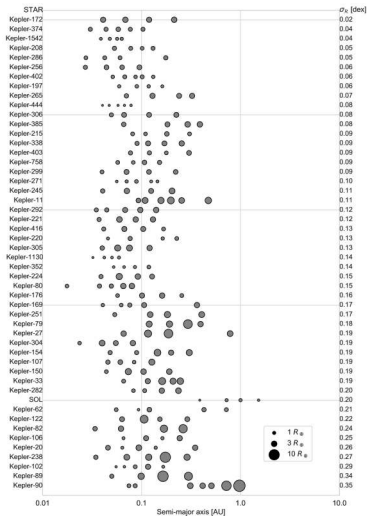
Transiting census; box and filled circles are compact multis (top), five uniform ones (middle), three non-uniform (bottom). Compact multi-planet systems exhibit striking architectural uniformity. Weiss et al. (2023, PPVII).

Where the Solar System is Atypical

The architecture of the solar system diverges noticeably from typical planetary systems discovered by exoplanet surveys.

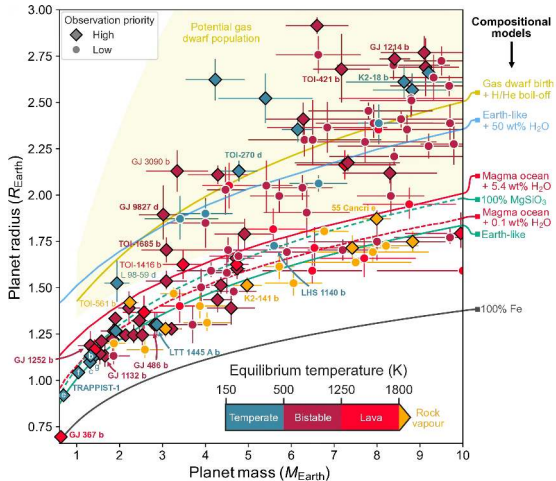
- No super-Earths or sub-Neptunes: the most common exoplanet size class is absent
- No close-in giant planets: hot Jupiters orbit $\sim 0.5\text{--}1\%$ of Sun-like stars, hot Neptunes are rarer still
- Outer giants on wide, circular orbits, while inner terrestrials lack compact uniformity

Peas-in-a-Pod: Solar System Doesn't Fit



- Compact multi-systems with ≥ 4 transiting planets within 1.52 AU.
- Ranked by uniformity of planet sizes.
- Solar system: *bottom quintile* of size uniformity.
- Inner solar system did not produce the typical compact-multi architecture.

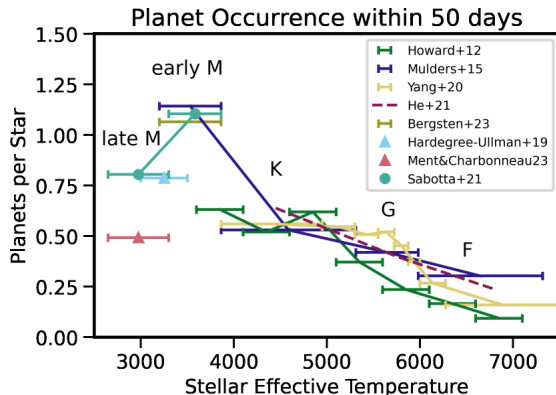
Mass-Radius: Where Each Planet Sits



- EOS tracks: 100% Fe, Earth-like, MgSiO_3 , water-rich, magma + H_2O , gas dwarf.
- Solar system terrestrials sit on the Earth-like curve.
- Close-in super-Earths sit there too: *stripped cores* that lost gas envelopes.
- Larger planets carry volatile envelopes that swell radii.

Source: Lichtenberg et al. (2025)

Occurrence Climbs Toward M Dwarfs



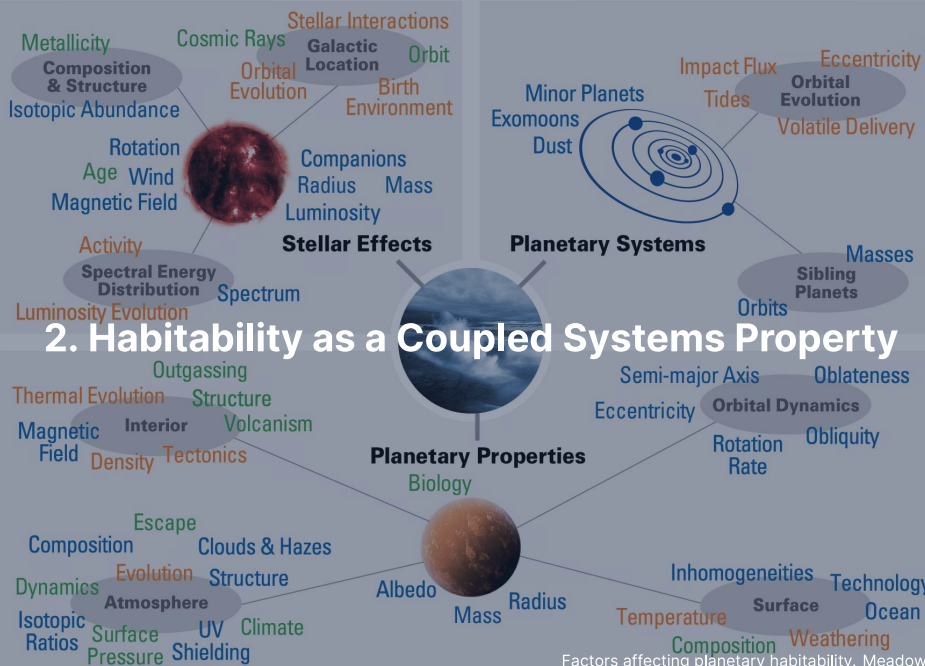
- Small planets ($1-4 R_{\oplus}$, $P < 50$ d) per star vs T_{eff} .
- Occurrence roughly $2\times$ higher around early M dwarfs than F dwarfs.
- Trend continues into late M dwarfs.
- Earth-sized planets are common around the most numerous stars.

Source: Mulders (2024)

Is the Solar System Rare?

Whether our architecture is rare remains open because 1 AU terrestrial analogues sit at current survey detection limits.

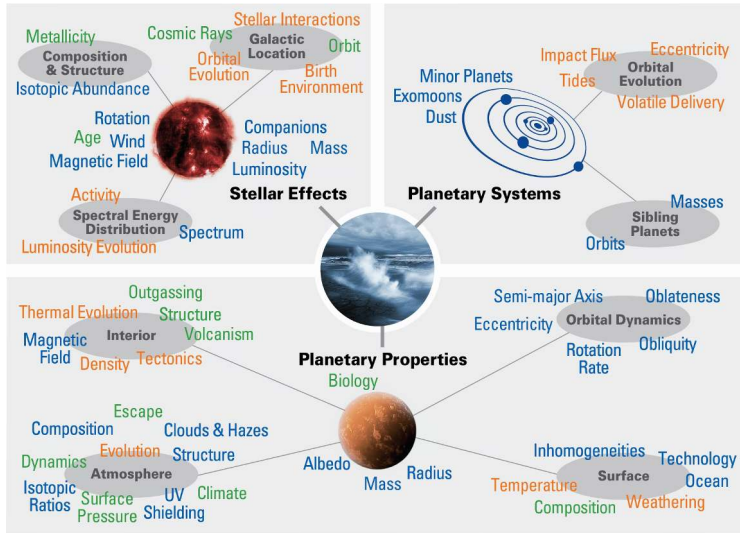
- Kepler four-year baseline limited sensitivity to long-period orbital architectures
- Inner solar system is structurally atypical compared to known compact multi systems
- PLATO (2027) and Gaia DR4 (late 2026) will resolve true occurrence



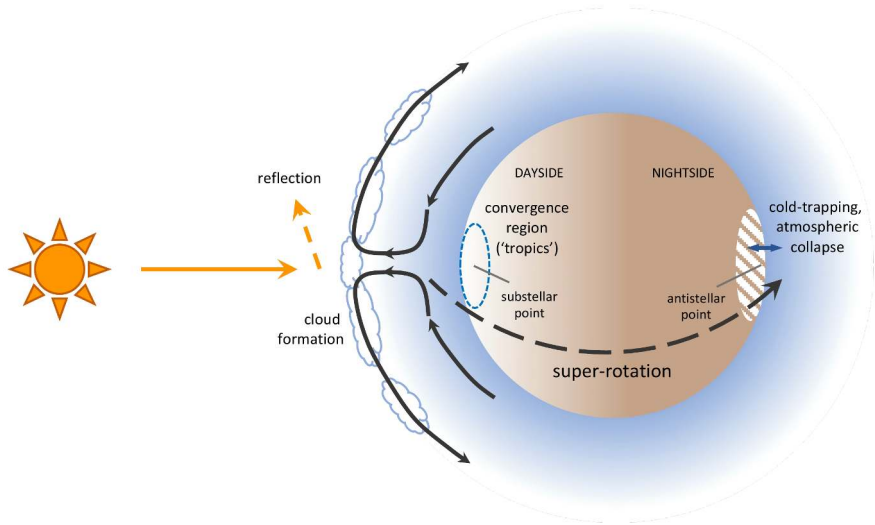
The Stack of Requirements

Habitability is not an isolated checklist but a coupled stack of astrophysical, geophysical, and geochemical conditions.

- Stellar host: stable luminosity over Gyr, benign stellar activity, and orbit in habitable zone
- Geophysical engine: bulk mass driving internal dynamo and tectonic volatile recycling
- Geochemical envelope: atmospheric pressure, escape rate, and surface liquid water stability
- Biosphere: feeds back on weathering and atmospheric redox; every level couples to its neighbours in both directions

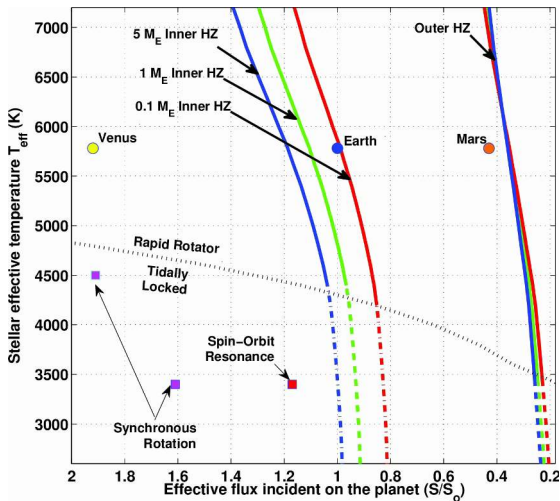


Factors affecting habitability: stellar, planetary, and orbital properties jointly shape the surface environment; breaking any coupling can terminate surface water stability. Blue: observable; green: model-interpreted; orange: theoretical. Meadows & Barnes (2018).



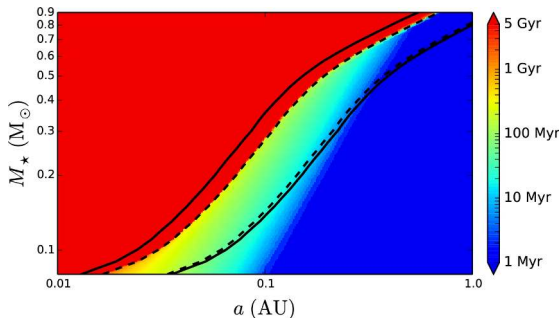
Atmospheric circulation on a tidally locked rocky planet: substellar convection and equatorial super-rotation transport heat, while nightside cold traps risk atmospheric collapse.
Wordsworth & Kreidberg (2022).

The HZ is Not a Line



- Inner edge: runaway greenhouse limit.
- Outer edge: maximum CO_2 greenhouse limit.
- Two planets at identical flux can have radically different climates.
- Time-dependent: stars brighten over main-sequence lifetime.
- Orbiting in the habitable zone today does not guarantee a habitable history.

M Dwarf HZ: Pre-MS Runaway Phase



- Duration of runaway greenhouse on a water-bearing planet vs $M_{\star}$.
- Late M dwarfs: up to a Gyr of runaway before reaching the main sequence.
- Planet starts *inside* runaway zone and loses initial water inventory.

Source: Luger & Barnes (2015)

3. Magma Oceans, Water Delivery, Tectonic Thermostat



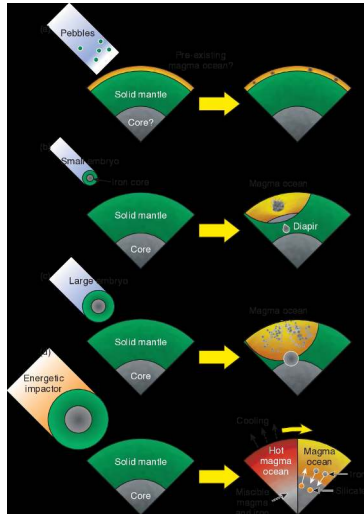
Magma-ocean differentiation. Lichtenberg et al. (2023)

Why "Delivery" Isn't the Right Frame

Terrestrial volatile inventories diverge because planetary differentiation and subsequent evolution matter far more than initial delivery.

- Earth, Venus, and Mars received similar initial accreted volatile budgets
- Magma ocean partitioning sets core, mantle, and steam atmosphere reservoirs
- Evolutionary controls: early hydrodynamic escape, mantle outgassing, and stellar brightening

Magma Ocean Differentiation Sets Boundary Conditions



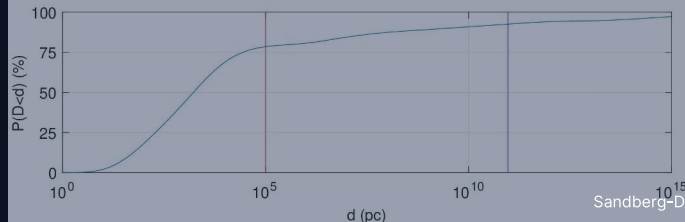
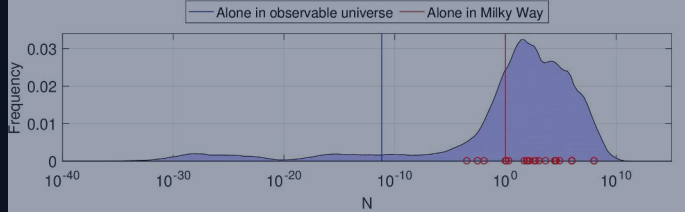
- Initially molten mantle separates from sinking metallic iron core.
- Volatiles partition between liquid silicate and steam atmosphere.
- Sequestered fraction depends on mantle redox, solidification time, and depth.
- Identical initial budgets yield radically different atmospheric masses.

Source: Lichtenberg et al. (2023)

Tectonic Regime: The Long-Term Thermostat

The carbonate-silicate cycle acts as a negative climate feedback that buffers surface temperature over geological timescales.

- Volcanic CO₂ outgassing balanced by temperature-dependent silicate weathering
- Subduction recycles carbonate minerals into mantle; failure produces Venus-like 92 bar
- Requires surface liquid water, active volcanism, and tectonic mantle recycling



4. Drake Equation and Fermi Paradox

The Drake Equation: A Framing Tool

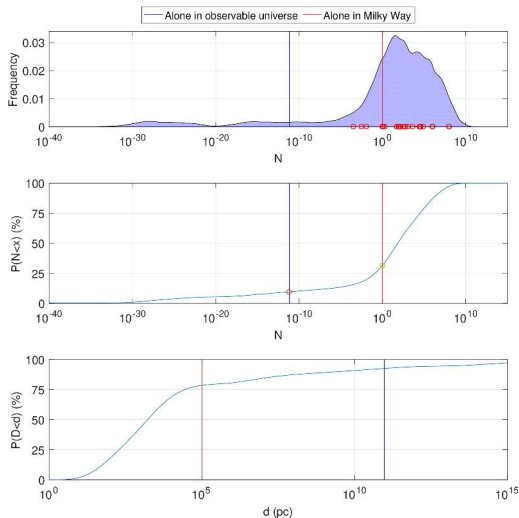
$$N = R_{\star} \cdot f_p \cdot n_e \cdot f_l \cdot f_i \cdot f_c \cdot L$$

$R_{\star}$ star-formation rate, f_p fraction with planets, n_e habitable planets per system, f_l life, f_i intelligence, f_c communication, L lifetime of a signalling civilisation.

- Astrophysical factors are now measured: $R_{\star}, f_p \sim 1, n_e \approx 0.1\text{--}0.6$ (Bryson+ 2021)
- Biological and social terms (f_l, f_i, f_c, L) span 4–10 orders of magnitude
- Factors co-evolve rather than acting as independent pipeline stages

The Drake equation is a tool for framing ignorance, not a numerical estimator.

Sandberg+ 2018: Honest Posterior Dissolves Fermi



- Monte Carlo sampling over full published parameter ranges.
- Resulting probability distribution for N is *bimodal*.
- Substantial probability mass sits below $N = 1$ in the Milky Way.
- Apparent cosmic silence is unsurprising once priors are honest.

Source: Sandberg, Drexler & Ord (2018)

Why Bimodal? Log-Sum of Broad Priors

Taking the logarithm turns the Drake product into a sum of broad priors, so $\log N$ is spread over many orders of magnitude.

$$\log_{10} N = \log_{10}(R_{\star} f_p n_e) + \log_{10} f_l + \log_{10} f_i + \log_{10} f_c + \log_{10} L$$

- Unconstrained factors have log-uniform priors spanning many decades
- Log-sum of broad, skewed priors: $\log N$ spreads over tens of decades, and the long low tail of f_l makes the result bimodal
- Posterior width spans ~ 30 decades, placing high probability on $N < 1$

Fermi Paradox: A Question, Not an Answer

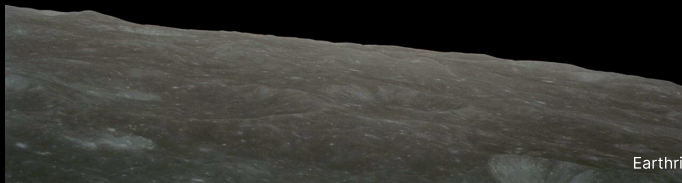
The apparent silence of the cosmos acts as an intuition check on subjective priors rather than physical evidence.

- Rare origin: abiogenesis ($f_i \ll 1$) or intelligence ($f_i \ll 1$) is exceptionally improbable
- Great filter: civilisations self-terminate rapidly, making lifetime L short
- Observational threshold: transmissions exist but lie below current detection limits

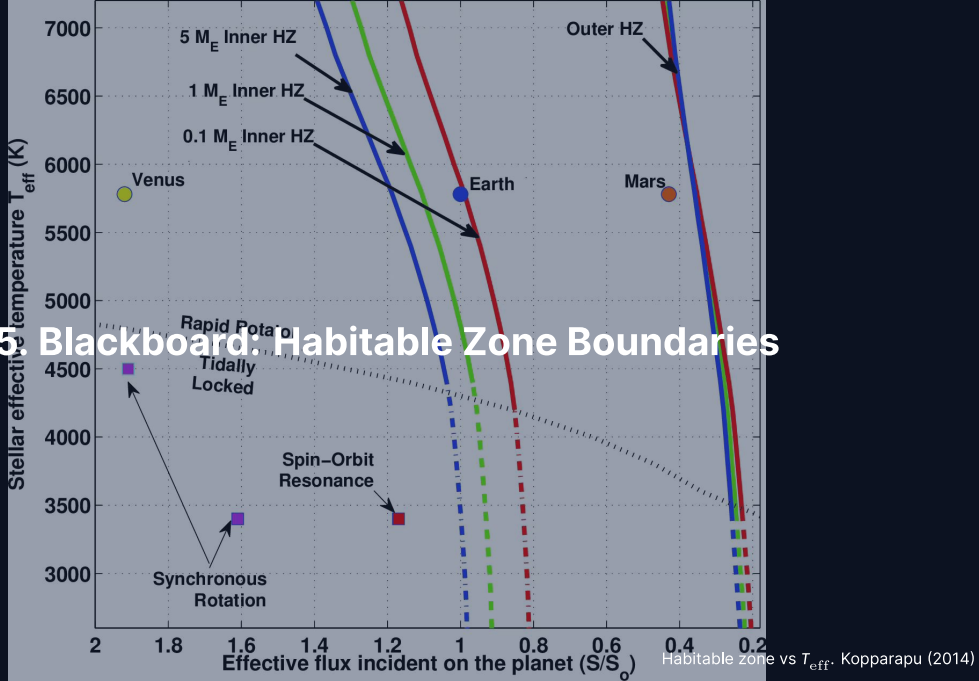
Break



End of Part 1: Solar System Context and Habitability
Part 2: Habitable Zone Derivation, Astrobiology, Life Detection



5. Blackboard: Habitable Zone Boundaries



Setup: Stellar Flux and Equilibrium Temperature

Star with bolometric luminosity $L_{\star}$, planet at distance d :

$$F_{\star}(d) = \frac{L_{\star}}{4\pi d^2}$$

Bond albedo A_B (reflected fraction), rapidly rotating sphere, equilibrium:

$$T_{\text{eq}}(d) = \left[\frac{L_{\star}(1 - A_B)}{16\pi\sigma\epsilon d^2} \right]^{1/4}$$

For Earth: $A_B = 0.3$, $\epsilon = 1$, $L_{\star} = L_{\odot}$, $d = 1 \text{ AU} \rightarrow T_{\text{eq}} \approx 255 \text{ K}$.

Surface is 288 K from $\sim 33 \text{ K}$ greenhouse warming. *To the board.*

Result: Habitable Zone Boundaries

Effective flux $S_{\text{eff}} = F_{\star}(d)/S_{\oplus}$ and the boundary distances:

$$d = 1 \text{ AU} \sqrt{\frac{L_{\star}/L_{\odot}}{S_{\text{eff}}}}$$

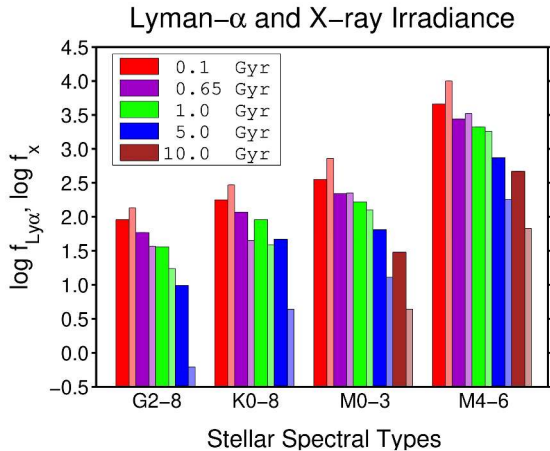
- Inner edge: runaway greenhouse (Simpson-Nakajima, Lecture 9), $S_{\text{in}} \approx 1.06$, ≈ 0.97 AU for the Sun
- Outer edge: maximum CO₂ greenhouse, $S_{\text{out}} \approx 0.35$, ~ 1.69 AU for the Sun
- Main-sequence lifetime $t_{\text{MS}} \approx 10 \text{ Gyr} (M_{\star}/M_{\odot})^{-2.5}$ sets how long the zone lasts

HZ Across Stellar Types

Habitable zone distance scales with stellar luminosity: $d \propto \sqrt{L_{\star}}$.

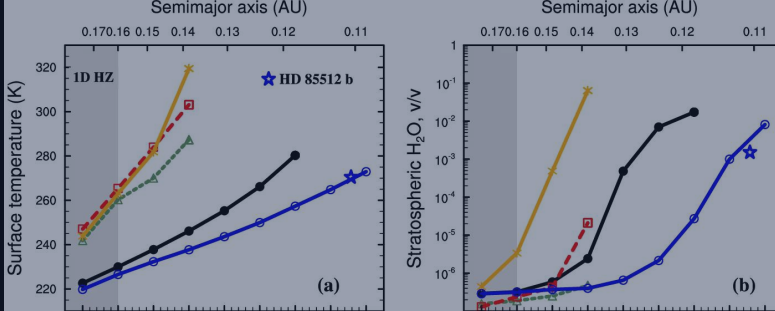
- **G dwarf** ($L_{\odot}$): $d = 0.97\text{--}1.69$ AU; main-sequence lifetime ~ 10 Gyr
- **K dwarf** ($0.3 L_{\odot}$): $d = 0.53\text{--}0.93$ AU; lifetime 24 Gyr (orange-dwarf sweet spot)
- **M dwarf** ($5 \times 10^{-4} L_{\odot}$): $d = 0.022\text{--}0.038$ AU; lifetime 10^{12} yr, but tidally locked with long pre-MS runaway

XUV at the HZ: M Dwarfs are Hostile

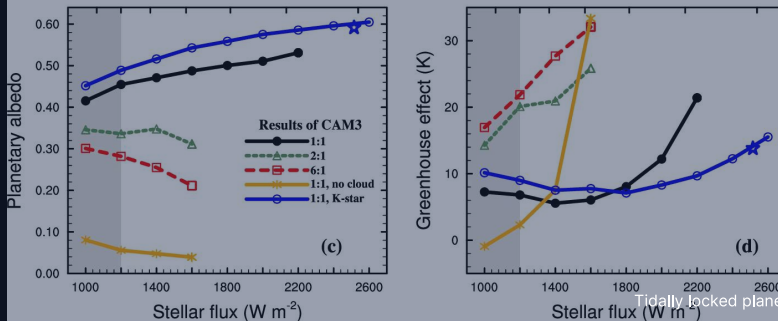


- Lyman- α and X-ray fluxes at Earth-equivalent distance.
- Flux is > 100–500 $\times$ higher around M dwarfs than G dwarfs.
- X-rays drop 500 $\times$ over 5 Gyr; Lyman- α drops only 5–10 $\times$.
- Early K dwarfs offer long lifetimes without M-dwarf XUV extremes.

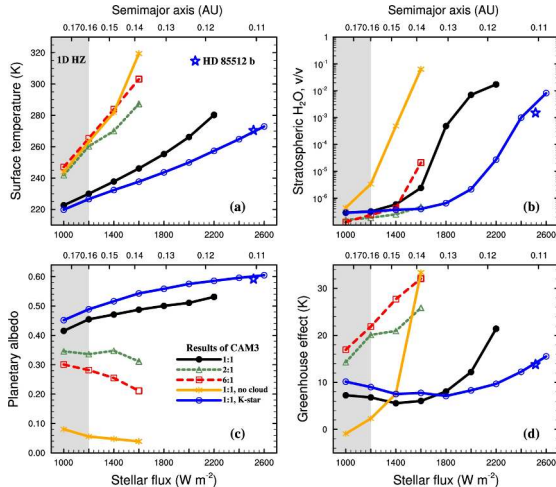
Source: Cuntz & Guinan (2016)



6. HZ Corrections: 3D Climate, Tidal Locking



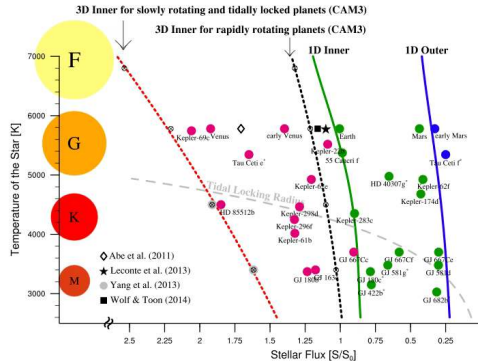
3D GCMs Shift Inner Edge of M Dwarf HZ



- 3D global climate model of tidally locked planet with clouds.
- Persistent substellar convection lifts thick convective water clouds.
- The planetary albedo rises to ~ 0.6 and reflects more stellar flux.
- Inner edge tolerates $\sim 2\times$ higher flux than 1D models predict.

Source: Yang et al. (2013)

Where M Dwarf HZ Planets Actually Sit



- Shields+ 2016; stellar flux increases toward the left.
- Solid curves: 1D Kopparapu inner and outer boundaries.
- Red dotted: 3D inner edge tolerates 1.7–2.2× 1D flux.
- Grey dashed: tidal-locking radius encloses almost all M-dwarf HZ planets.

Tidal-Locking Timescale

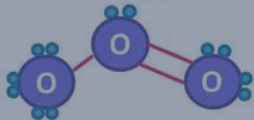
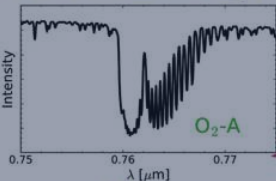
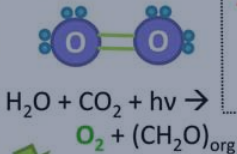
Tidal despinning scales steeply as a^6 , ensuring that close-in habitable planets around low-mass stars synchronize their rotation rapidly.

$$\tau_{\text{lock}} \sim \frac{\omega_0 a^6 I_p Q_p}{3 G M_{\star}^2 k_{2,p} R_p^5}$$

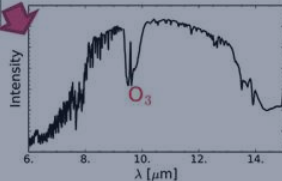
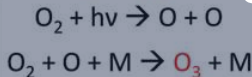
- Earth at 1 AU has $\tau_{\text{lock}} \sim 10^{12}$ yr, far exceeding universe age
- TRAPPIST-1 b ($a \approx 0.011$ AU) synchronises in $\lesssim 10^7$ yr
- With system age ~ 8 Gyr, all known M-dwarf HZ worlds are locked

Gaseous

Ex: **Oxygenic Photosynthesis**



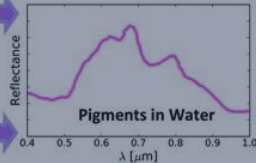
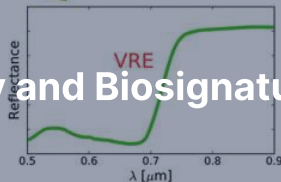
7. Astrobiology and Biosignatures



Surface



$$\text{NDVI} = \frac{\text{IR} - \text{VIS}}{\text{IR} + \text{VIS}}$$



Temporal



Seasonal changes in Gases or Surface



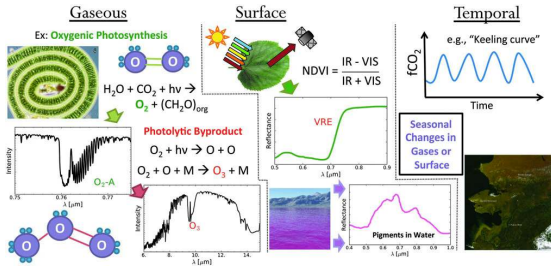
Defining Life and Its Origin Hypotheses

Working definition (Lederberg 1965): a self-replicating, metabolising, evolving chemical system.

Four origin-of-life hypotheses:

- **RNA world:** catalytic ribozymes carry the genetic code before proteins
- **Alkaline hydrothermal vents:** geochemical proton gradients drive early metabolism
- **Warm surface ponds:** wet-dry cycles polymerise organic precursors
- **Panspermia:** transfer of viable organisms between bodies

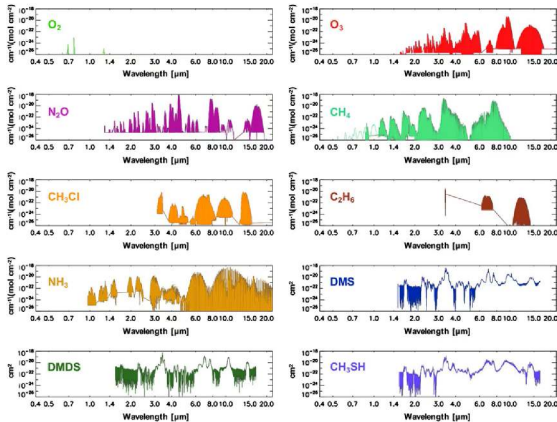
Three Classes of Biosignature



- **Gaseous:** atmospheric O_2 , O_3 , CH_4 , N_2O , and organosulfur species.
- **Surface:** vegetation red edge with sharp reflectance jump near 700 nm.
- **Temporal:** seasonal cycles like Earth's CO_2 Keeling oscillation.
- A credible detection needs several classes together.

Source: Schwieterman et al. (2018)

Biosignature Gases in the Mid-IR



- Ten candidate biosignature gases display characteristic mid-IR bands.
- Key features: O₃ at 9.6 μm, CH₄ at 3.3 and 7.7 μm.
- N₂O features and DMS / DMDS window in 3–15 μm.
- These bands set the wavelength needs of LIFE (mid-IR interferometer) and HWO (Habitable Worlds Observatory).

Source: Schwieterman et al. (2018)

Disequilibrium is the Diagnostic

Chemically incompatible gases coexisting in an atmosphere provide the most convincing remote signature of active biological production.

- Individual gases in isolation almost always possess plausible abiotic origins
- Earth pair: O_2 (21%) and CH_4 (1.8 ppm) react on decade timescales
- Simultaneous persistence requires massive, continuous biological resupply fluxes

Catling 2018: A Bayesian Framework

EXOPLANET BIOSIGNATURES ASSESSMENT

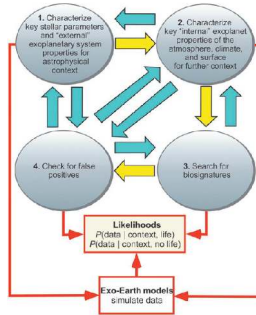
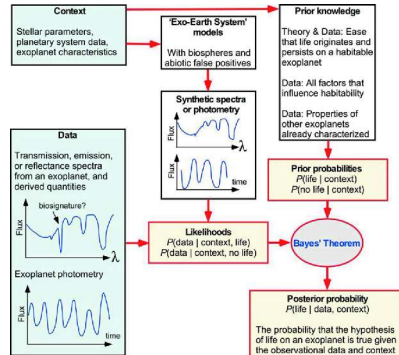


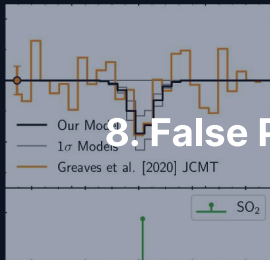
FIG. 2. Exo-Earth models to assess whether external



Detection requires a Bayesian chain of inference incorporating stellar context, planetary properties, and abiotic mechanisms, not a single yes/no spectral peak.

Source: Catling et al. (2018)

8. False Positives and Solar System Targets



False Positives: Every Gas Has an Abiotic Pathway

Biosignature verification is fundamentally an inverse problem because abiotic mechanisms can produce every individual candidate gas.

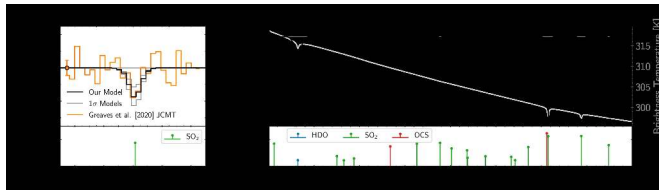
- Abiotic O_2 : H_2O photolysis followed by hydrogen escape, or CO_2 photolysis
- Abiotic CH_4 : hydrothermal serpentinisation or volcanic outgassing at low oxygen fugacity
- Abiotic N_2O from lightning and photochemistry; DMS has no significant abiotic Earth source, but one could exist elsewhere
- Confirmation requires ruling out all plausible geochemical and photochemical false positives

Solar System Targets for Life Detection

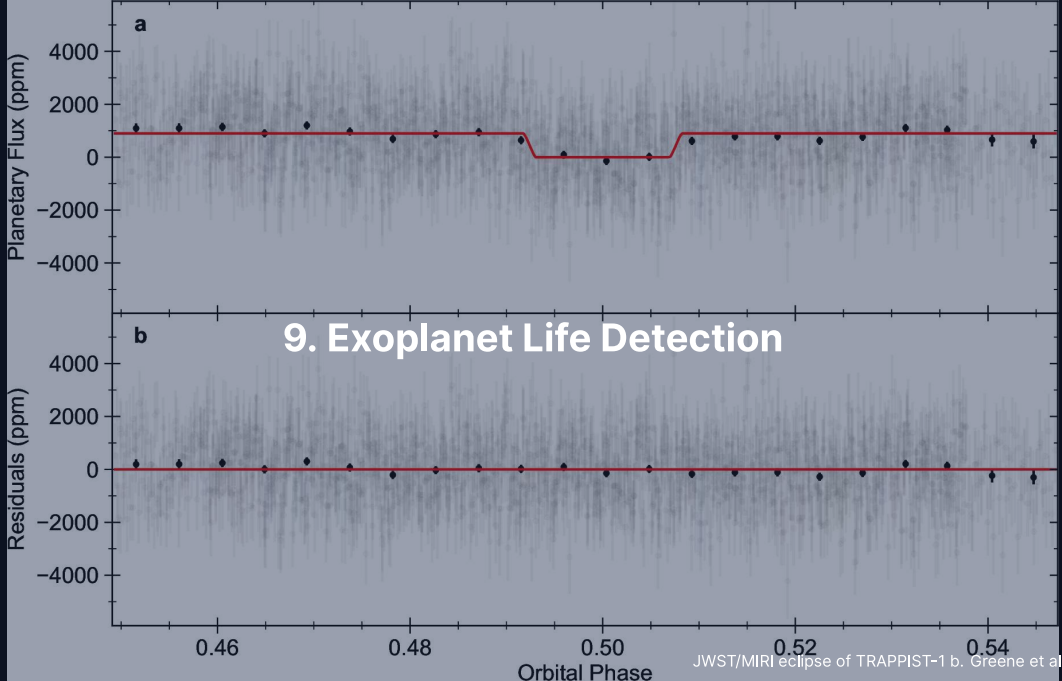
In situ solar system exploration targets five prime environments with liquid water reservoirs or rich organic chemical syntheses.

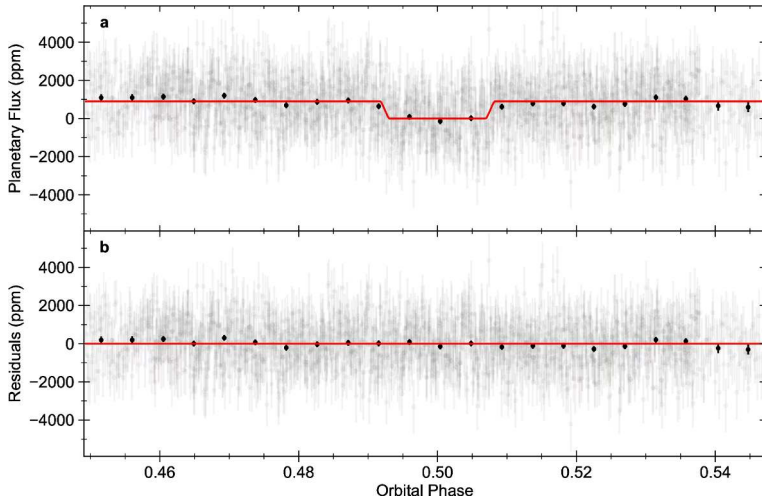
- Mars: ancient habitable lacustrine environments; Mars Sample Return path
- Europa and Enceladus: confirmed subsurface oceans; Enceladus's plume carries H_2 , organics and phosphates from hydrothermal activity, hypothesised for Europa
- Titan and Venus: complex hydrocarbon hydrology and contested cloud chemistry

Venus Phosphine: A Cautionary Case Study



- Greaves et al. (2021): PH₃ in the Venus clouds, a biosignature?
- Lincowski et al. (2021): the line is fit by mesospheric SO₂
- K2-18 b, same lesson: CH₄ (5σ) and CO₂ (3σ) stand; DMS fell from 2.4σ to ~ 1σ
- DAVINCI tests this in situ, EnVision and VERITAS from orbit



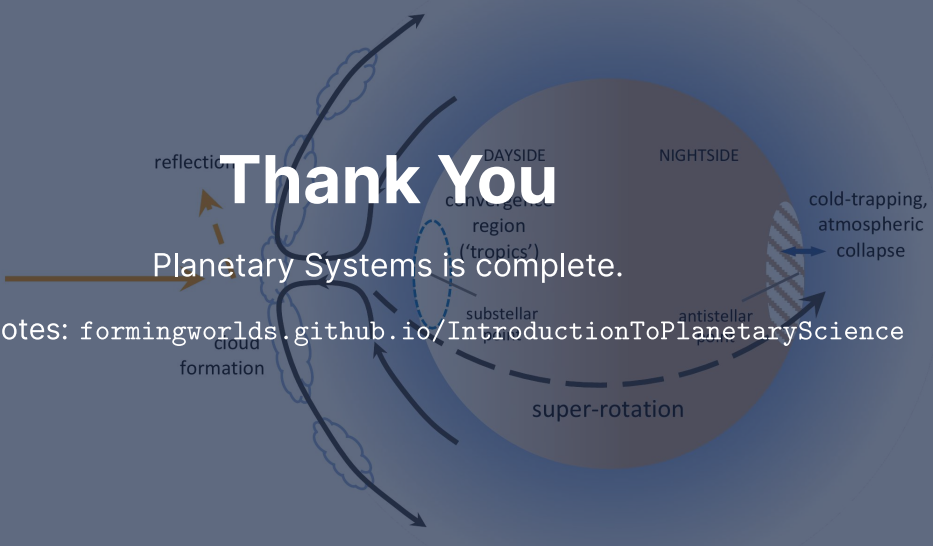


JWST MIRI $15\,\mu\text{m}$ secondary eclipse of TRAPPIST-1 b measures a dayside brightness temperature of 503 K, matching the 508 K bare-rock prediction and ruling out a thick atmosphere. Credit: Greene et al. (2023).

Summary

1. Planet formation and differentiation are universal physical processes
2. Interiors are heat engines: tectonics, dynamos and outgassing over Gyr
3. Atmospheres evolve through escape, outgassing and photochemistry
4. Habitability is a coupled trajectory, not a static orbital condition
5. The solar system is one detailed reference in a diverse exoplanet population

Next: The final exam.



A diagram of a planet with various atmospheric features labeled. On the left, a sun icon emits a horizontal arrow towards the planet. The planet is divided into a 'DAYSIDE' (left) and a 'NIGHTSIDE' (right). On the dayside, there is a 'convergence region ('tropics')' marked with a dashed blue circle, and 'cloud formation' is indicated by blue cloud shapes. A 'reflection' arrow points away from the dayside. On the nightside, 'cold-trapping, atmospheric collapse' is shown in a hatched region, and 'super-rotation' is indicated by a dashed curved arrow. Other labels include 'substellar point' and 'antistellar point'.

Thank You

Planetary Systems is complete.

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience